

Use Case 1: Prior Authorization Decision Support

Turn a slow stack of faxed clinical documents into a clear, cited coverage decision in minutes, with a human in charge of every hard call.

Business area: Utilization Management (UM)

This is a self-contained brief. Everything you need is in this file.

Why this role matters: Prior Authorization (PA) is a high-volume, heavily regulated program that sits at the center of the payer and provider relationship. It is a strategic, enterprise-priority effort with senior leadership visibility. It is the most common friction point between payers and providers, and every delayed decision is a member waiting on care. Getting PA right improves provider trust, lowers administrative cost, and gets members to needed care sooner. That is why this work carries weight across the organization.

The problem

Prior Authorization is the approval a provider must get from the plan before a service, procedure, or drug is covered. Today most of that work is manual. Requests arrive by fax or portal, packed with clinical documents that vary in format and quality. Someone has to open each one, find the relevant facts, and figure out whether the request is complete.

UM nurses and medical directors then review each request against medical policy and clinical rules. They confirm the diagnosis, the history, the tried-and-failed treatments, and the specific criteria that a service requires. This is careful work, and it is repetitive. The clear-cut cases take almost as long as the hard ones because the reading and matching still has to happen by hand.

The results are slow turnaround, heavy administrative load for both the provider and the plan, and decisions that can vary from reviewer to reviewer. Members feel the stress of waiting. Part of the reason is that the information needed for a single decision is spread across many places: benefits, eligibility, clinical policy, and provider systems. No one view holds the full picture, so people spend time hunting instead of deciding.

Scenario (illustrative, made-up)

John Doe, age 71, is scheduled for an elective open lumbar spinal fusion. The hospital's utilization team submits a prior authorization request for an inpatient admission and faxes a 14-page packet: the surgeon's operative plan, a history and physical, a cardiology clearance, and the medication list.

A UM nurse opens the fax. The medical policy for an inpatient admission requires that the case meet criteria for inpatient level of care rather than outpatient or observation: the procedure is appropriate for an inpatient setting, there are documented comorbidities that raise surgical risk, the expected stay crosses at least two midnights, and there are post-operative care needs that cannot be safely managed on an outpatient basis. The nurse reads all 14 pages to find these facts. The comorbidities

are in the history and physical, the cardiology clearance is a separate page, but the expected length of stay is never clearly stated. The nurse pends the case for more information. The admission date is at risk while the hospital and the plan trade faxes over a detail a cleaner read of the same packet could have resolved in minutes.

Who it is for

- **Intake team:** the people who receive requests and check them for completeness.
- **UM nurses and medical directors:** the clinical reviewers who apply policy and make determinations.
- **Providers and their staff:** the offices that submit requests and chase status.
- Note: members are the ultimate beneficiary, since faster, more consistent decisions get them to care sooner.

What a weak idea looks like

- A chatbot that answers typed questions with generic policy text, ignores the actual clinical documents, and lives outside the decision workflow. It makes a reviewer type in what the AI should have read. That is a wrapper, and it moves no real work.
- Not capturing the experience and the workflow. Strong AI wrapped in a bad experience still leads to a bad product. If you do not capture the workflow, the steps, the hand-offs, and what the user does, the AI capability alone will not carry it. The workflow elements are as important as the AI capabilities.

What to deliver

Produce the following seven items.

1. **Problem Brief.** A short, sharp statement of the problem, the user you chose, and why now.
2. **Service Design.** The current end-to-end journey and the future journey, showing where the AI changes the flow.
3. **Working Prototype.** An interactive demo built with a tool such as v0, Claude Code, Cursor, Replit, Lovable, Codex, or similar. It does not need to be production ready.
4. **Product Roadmap.** A phased plan, with the reasoning for the order.
5. **Key Tradeoff.** One real tradeoff you faced, the options, and why you chose as you did.
6. **Business Case, Value Proposition, and ROI.** A clear description of the quantifiable value this drives, using placeholder numbers. Cover the value proposition for each stakeholder (the member, the provider, the UM nurse and medical director, and the plan), what it takes to build and run, and an ROI formulation that names its key drivers. Remember the logic of the ROI needs to make sense more than the actual numbers, and think beyond admin savings.
7. **Evals Framework.** A clear AI evaluation framework that showcases your approach even if you have not run it with data. Define what a good answer looks like, build a small set of test cases with expected outcomes, and show how you would measure the eval metrics. For a regulated use case, evals are how we learn to trust the AI to do real work.

Narrate through a persona: Assume clear personas for this use case and pick one as the main character of your demo. Walk us through the prototype in that person's lens, step by step. Show what they see and do today, before this AI tool, and then what they see and do with it. Make the contrast concrete so the benefit we would gain is obvious, not asserted.

Show the real workflow: Any specific workflow elements that need to be captured for this process should be showcased clearly in the prototype. Do not abstract them away. If a step, decision point, hand-off, or required input matters to how the work actually gets done, make it visible in the demo so we can see the process, not just the outcome.

Cover the end-to-end journey: The prototype must cover the full end-to-end journey, not just small aspects of it. We want to see the whole flow from start to finish, how each step connects to the next, and where the AI changes the work along the way. A demo that polishes one screen while skipping the rest of the journey does not show us enough.

Sample data you can use (made-up)

- **Member:** John Doe, age 71, member ID MBR-0042 (illustrative).
- **Requested service:** Elective inpatient admission for open lumbar spinal fusion, L4 to L5 (illustrative).
- **Medical policy (illustrative), an inpatient admission requires inpatient level of care, meaning:**
 - The planned procedure is appropriate for an inpatient setting and is not on the outpatient-only list.
 - Documented comorbidities that raise peri-operative risk (for example heart failure, COPD, or uncontrolled diabetes).
 - An expected length of stay crossing at least two midnights.
 - Post-operative monitoring or care needs that cannot be safely provided in an outpatient or observation setting.
- **Clinical note (illustrative, partially meets criteria):** "71-year-old scheduled for open lumbar spinal fusion at L4 to L5. History of heart failure (EF 40 percent) and type 2 diabetes. Cardiology clearance on file. Surgeon expects a multi-day stay for pain control and mobilization. Exact length of stay not documented." (Meets the comorbidity and setting criteria. The expected length of stay is implied but not clearly documented, so completeness is uncertain.)

The mindset and experience we are looking for

This role fits someone who works the way this program is being built. We are looking for the sum of a person's experience showing up in how they approach the problem, not a blank-slate attempt. Signals that matter:

- A builder who prototypes to think. You would rather show a working thing in two days than write a spec for two weeks, and you learn by building.

- Architecture-first instincts. You separate content from workflow, treat knowledge as a single governed source of truth, and design so one thing feeds many channels instead of creating silos.
- Real experience in regulated, high-volume operations. You respect medical policy and compliance, and you know exactly where a human must stay in the loop.
- Reasons from the frontline pain. You start from the member and the provider, not from the technology.
- Connects the piece to the whole. You see how one use case fits the larger platform and sequence your roadmap accordingly.

If that describes how you already work, this brief should feel natural. Show us your reasoning and let your past experience culminate in the approach.

Show your thinking

We are hiring for judgment, not just output. Alongside what you build, make your reasoning visible. The follow-up call will go deep here.

- The decisions behind the work. What you considered, what you chose, and what you deliberately did not build, and why.
- Your assumptions. What you assumed about the members, the providers, the clinical data, and the constraints.
- How your experience shaped this. Point to one or two things from your past work that directly influenced your approach. We read the strongest submissions as the sum of everything a person has learned, applied to this problem.
- What you would do next. With more time, and what you would do differently knowing what you know now.

A rough prototype with sharp, well-argued thinking beats a polished demo you cannot defend.

How we will evaluate

Dimension	What we are checking
Problem & user insight	Real pain, real users, a clear "why now"
AI that does real work	AI cannot be just a wrapper; it needs to drive significant portion of the value
Journey mapping	Current, future, how the product gets there
Execution	The prototype shows the core experience convincingly
Judgment	Roadmap order and the chosen tradeoff
Responsible AI	Explicit, sensible human-in-the-loop for regulated moments

Thinking & accumulated judgment	Clear reasoning, considered alternatives, and prior experience that visibly shapes the approach, weighted as heavily as the prototype
Business case	A clear description of the quantifiable value we can drive using placeholder numbers. Remember the logic of the ROI needs to make sense more than the actual numbers. Think beyond admin savings.
Evals	A clear AI Eval framework that showcases your approach even if you have not run it with data

Guidance and rules

You have 2 to 3 days. Use made-up data only, no real member, provider, or claims data.

Keep a human in the loop where it counts. Clinical and coverage decisions are regulated and serious. Be ready to present. The follow-up is a working conversation: we want the reasoning behind each choice, the paths you did not take, and how your experience led you here.

How cases are decided: A prior authorization determination is grounded in published coverage and clinical guidelines, not personal opinion. Reviewers apply National Coverage Determinations (NCD), Local Coverage Determinations (LCD), Humana internal medical policy, and MCG guidelines in a hierarchical manner to approve or deny a case. Your solution should ground the AI in these same sources and make clear which guideline drove each recommendation.

Use Case 2: Agent Assist for Advocates

Every regulated call, guided as it happens — the right words, the member’s own data, the next best step, and zero after-call scramble.

Business area: Contact Center — Member Services & Pharmacy Operations

Main AI role: A real-time co-pilot on the live call that understands what is happening, surfaces exactly what to say and do next, keeps the advocate compliant in the moment, and does the after-call work.

This is a self-contained brief. Everything you need is in this file.

Why this role matters: Agent Assist is one of the most critical initiatives at Humana. It sits on the highest-volume, most heavily regulated member touchpoint — the live call — and it is the proving ground for how AI and human advocates work together across every line of business. Done right, it lifts compliance, quality, and member experience at the same time. This is a strategic, enterprise-priority program with senior visibility, not a peripheral experiment. Whoever we hire will be building at the center of it.

The problem

Advocates handle regulated, multi-step calls — education, enrollment, refills, pricing, do-not-call. On a single call they must say mandatory legal verbatim word-for-word, pull the member’s context from several systems, follow the correct flow for that call type, handle objections, and document the outcome accurately — all while talking to a live person. Today they do this by hunting through dense knowledge documents, switching between system tabs, and leaning on memory and tenure.

The result: missed or paraphrased legal verbatim (a real compliance risk), inconsistent answers, long handle times, heavy after-call work, and months-long ramp for new hires. The information needed is spread across benefits, eligibility, claims, pharmacy, provider, and scripting systems.

Scenario (illustrative, made-up). Harry calls about a refill. Mid-call it turns into an education-and-enrollment opportunity. The advocate must deliver three pieces of legal verbatim exactly, quote Harry’s real drug costs, secure an “absolute yes,” and log a clean disposition — without losing him. One missed verbatim is a compliance finding. One fumbled tab is four seconds of dead air.

Why now

Call volume and regulatory scrutiny are rising while average tenure is falling. QA samples only a small fraction of calls, so misses surface weeks later instead of being prevented. And AI is finally good enough to follow a live conversation, ground answers in the member’s own data, and enforce verbatim in the moment. *(Any figures you cite may be illustrative — label them as such.)*

Who it is for (pick your main user)

- Contact center advocates
- Team leads / QA & compliance

What great looks like (and what weak looks like)

Great: a live call where the AI detects the intent and call type, pulls the member's (fake) data, surfaces the exact required verbatim at the right beat, flags a missed statement in real time, and auto-drafts the wrap and warm handoff. AI clearly changes the outcome.

Weak: a chatbot side-panel that answers typed questions with generic content and ignores the call flow, the compliance requirement, and the after-call work. A wrapper, not a co-pilot.

What to deliver

Build an AI-powered prototype and turn in five things.

1. Problem Brief. Define the problem, who it is for, and why it is worth solving now. Call out the current pain and the business opportunity.

2. Service Design. Current end-to-end advocate journey (before, during, after the call) and the future journey. Make the compliance and handle-time pain clear and the improvement obvious.

3. Working Prototype. An interactive demo (v0, Claude Code, Cursor, Replit, Lovable, Codex, or similar). Not production-ready. It **must show**: (a) live call context, (b) the correct verbatim surfaced in-flow, (c) a real-time compliance nudge when something is missed, and (d) an auto-generated wrap / handoff.

4. Product Roadmap. A phased plan. For each phase, explain the order and the reason.

5. Key Tradeoff. Name one real tradeoff you made and explain why.

Where AI decides, recommends, or hands off

AI decides (autonomous)	AI recommends (human confirms)	Human only (AI hands off)
Surface the next step; detect a missed or paraphrased verbatim; draft the wrap note and disposition	Objection rebuttals; next-best-action; disposition code; when to warm-transfer	Commit an enrollment; make a coverage determination; take payment; give clinical advice

Sample data you can use (made-up)

- **Member:** "Harry Whitfield," Humana MAPD, preferred cost-share. Meds: atorvastatin 20mg, metformin 500mg. Fake retail vs. CenterWell prices.
- **Three required verbatim:** greeting on a recorded line; the pricing disclaimer; the "your decision today has no impact on your plan membership" closing.
- **Flow:** educate → confirm interest (absolute yes) → price → enroll or decline → wrap.

The mindset and experience we are looking for

This role fits someone who works the way this program is being built. We are looking for the sum of a person’s experience showing up in how they approach the problem — not a blank-slate attempt. Signals that matter:

- **A builder who prototypes to think.** You would rather show a working thing in two days than write a spec for two weeks, and you learn by building.
- **Architecture-first instincts.** You separate content from workflow, treat knowledge as a single governed source of truth, and design so one thing feeds many channels instead of creating silos.
- **Real experience in regulated, high-volume operations.** You respect legal verbatim and compliance, and you know exactly where a human must stay in the loop.
- **Reasons from the frontline pain.** You start from the member and the advocate, not from the technology.
- **Connects the piece to the whole.** You see how one use case fits the larger platform and sequence your roadmap accordingly.

If that describes how you already work, this brief should feel natural — show us your reasoning and let your past experience culminate in the approach.

Show your thinking (weighted as heavily as the prototype)

We are hiring for judgment, not just output. Alongside what you build, make your reasoning visible — the follow-up call will go deep here.

- **The decisions behind the work.** What you considered, what you chose, and what you deliberately did *not* build — and why.
- **Your assumptions.** What you assumed about the members, the advocates, the data, and the constraints.
- **How your experience shaped this.** Point to one or two things from your past work that directly influenced your approach. We read the strongest submissions as the sum of everything a person has learned, applied to this problem.
- **What you would do next.** With more time — and what you would do differently knowing what you know now.

A rough prototype with sharp, well-argued thinking beats a polished demo you cannot defend.

How we will evaluate

Dimension	What we are checking
Problem & user insight	Real pain, real users, a clear “why now”
AI that does real work	Live context, grounding, in-the-moment verbatim/compliance, automation — not a wrapper
Journey mapping	Current → future → how the product gets there

Execution	The prototype shows the core experience convincingly
Judgment	Roadmap order and the chosen tradeoff
Responsible AI	Explicit, sensible human-in-the-loop for regulated moments
Thinking & accumulated judgment	Clear reasoning, considered alternatives, and prior experience that visibly shapes the approach — weighted as heavily as the prototype

Guidance and rules

You have 2 to 3 days. Use made-up data only — no real member, provider, or claims data.

Keep a human in the loop where it matters. Clinical, coverage, and payment decisions are regulated and serious. The AI guides; the advocate speaks and decides. The AI must never auto-commit an enrollment, coverage determination, or payment. Mandatory legal verbatim is owned and approved by Legal — the AI surfaces it, it does not invent it.

Be ready to present. The follow-up is a working conversation: we want the reasoning behind each choice, the paths you did not take, and how your experience led you here.

Stretch (optional, for exceptional candidates)

Show the same knowledge that guided the advocate also driving the post-call QA score — guided text equals graded text — or estimate the impact on new-hire ramp time.

Use Case 3: Knowledge & Search

One source of truth for AI and humans. Author knowledge once, and each channel and each search gives the same correct, cited, current answer.

Business area: Knowledge Management, Content Operations & Search (serving every channel)

Main AI role: Turn a sprawl of unstructured documents into governed, AI-ready knowledge that each channel consumes, and keep it correct with AI-assisted ingestion, retrieval, and drift detection. Then make it findable: search that covers only the knowledge bases the asker may read, cites the unit it answers from, and logs the questions it could not answer so authors can fill the gaps.

Everything you need is in this file.

Why this role matters: Humana is building voice, chat, advocate assist, outbound, and digital on the same knowledge, and each is only as correct and as compliant as that knowledge. Search is how an advocate at the desk and an assistant at answer time reach it. The Knowledge & Search Platform is the layer under all of them: an enterprise priority with senior visibility, and the dependency the other AI use cases in this series wait on. You would be building that layer.

The problem

The answers members and advocates rely on live inside hundreds of unclassified, non-AI-ready documents: knowledge articles, benefit documents, scripts, policies. The same legal verbatim and policy language is duplicated across many documents, drifts over time, and is hard to keep current. An assistant retrieving over these raw documents returns inconsistent and sometimes non-compliant answers, because no one has classified the source. Mandatory legal verbatim is buried in long documents a human must open and follow by hand. There is no single source of truth, so each channel team writes its own answer.

Finding has the same problem. An advocate searches several tools, gets documents rather than answers, and gets different documents from each tool. Search cannot tell them which result is current or which wording Legal approved. No tool logs the questions that came back empty, so knowledge owners cannot see what advocates need and do not have.

Scenario (illustrative, made-up). The exact legal verbatim an advocate must read during an education call lives in one document; a slightly different wording of the “same” disclaimer lives in two others. The voice bot cites one, the advocate reads another, the website shows a third. When Legal revises the language, no one can find and update all the places it lives. Search makes the failure visible: type “Rx pricing disclaimer” today and you get three documents with three wordings and no way to tell which is current. This platform ends that failure. The same search returns one approved unit, with its citation and the channels that use it, and when Legal revises the wording, each channel’s search picks up the new unit at once.

Why now

Humana is building AI across its channels, and each one needs grounded, current, citable knowledge. You cannot retrieve your way out of an unstructured document pile. The knowledge layer is the

dependency under everything else. Search over a body of knowledge follows close behind the knowledge itself, so design the two together. AI is now good enough to help build and maintain it: extracting structured knowledge from documents, deduplicating, and flagging drift, with humans approving. *(Any figures you cite may be illustrative. Label them as such.)*

Who it is for (pick your main user)

- Knowledge Operations / content authors and Legal / Compliance approvers (author and govern)
- Advocates and assistants who search or ask, and the downstream channels and AI features that need grounded answers (the consumers)
- Knowledge owners who read what people searched for and could not find (the ones who close gaps)

What great looks like (and what weak looks like)

Great: a demo where the AI turns a messy source document into structured, tagged, cited knowledge units and detects duplicates; a consumer (assistant or advocate) searches or asks and gets an answer back *with a citation* to the unit; the same question from two roles returns answers scoped to each role; a question the knowledge cannot answer says so and lands in the author's queue; and the system flags a unit as stale when its source changes. AI does the extraction, matching, and flagging; humans approve.

Weak: a search box over the raw files, however fast. Also weak: search that answers from documents no approver has seen, or that cannot say when it has no answer. We are testing structure, governance, and single-source reuse. Search counts as retrieval over approved units and as the log of questions authors still need to answer.

What to deliver

Build an AI-powered prototype and turn in five things.

1. Problem Brief. Define the problem, who it is for, and why it is worth solving now. Call out the current pain and the business opportunity.

2. Service Design. The current end-to-end knowledge journey (how a policy or script goes from a source document to something an advocate or assistant uses today) and the future journey. Make the current pain clear and the improvement obvious. Include the finding journey: how an advocate or an assistant locates an answer today, and how they will find it once the platform exists.

3. Working Prototype. An interactive demo (v0, Claude Code, Cursor, Replit, Lovable, Codex, or similar). Not production-ready. It **must show**: (a) a document turned into structured knowledge units, (b) authoring and a governance / approval step, and (c) a consumer searching or asking and pulling a knowledge unit back with a citation to source, scoped to the knowledge bases that consumer's role may read. Optional (d): a question the knowledge cannot answer, which the prototype turns into a proposed unit for an author to write.

4. Product Roadmap. A phased plan. For each phase, explain the order and the reason. Put search on the plan: when it goes live relative to the knowledge it runs on, and why.

5. Key Tradeoff. Name one real tradeoff you made and explain why.

Where AI decides, recommends, or hands off

AI decides (autonomous)	AI recommends (human confirms)	Human only (AI hands off)
Extract units from a document; classify type; deduplicate; embed for retrieval; flag drift / expiry	Proposed wording updates; suggested tags and domain; merge/split suggestions	Approve or publish legal or clinical content; retire the source of record; resolve a conflict of authority
Blend and rank retrieval methods for a query; scope results to the knowledge bases the asker may read; decline when evidence is insufficient; cluster the questions that returned no answer	Proposed new units or question-and-answer pairs from unanswered questions; synonyms and acronyms learned from search logs (Humana-speak); a pinned answer for a top question	Grant a role access to a knowledge base; approve a unit written to close a gap; decide whether a top question deserves a written answer or a fix to the process that causes it

Sample data you can use (made-up)

- **Source doc:** a fake “Rx Education Script” with a greeting, a pricing disclaimer, and a closing, with some lines marked “Verbatim (Legal).”
- **A near-duplicate:** a second script whose disclaimer wording differs slightly (to test dedup and drift).
- **Consumers:** an advocate guidance view and a member chat answer, both pulling from the same unit with a citation.
- **A search log:** about 20 made-up questions advocates might type, including Humana-speak and acronyms, a few misspellings, and several the two scripts cannot answer (to test gap capture).
- **Two roles:** a Pharmacy advocate and an Insurance advocate with different knowledge-base access (to test scoping). Add a member-chat persona if you attempt the stretch.
- **A small golden set:** 5 to 10 questions with the expected answer and the unit each should cite (to show how you would measure retrieval).

The mindset and experience we are looking for

This role fits someone who works the way we are building this program. We want to see the sum of a person’s experience in how they approach the problem. Signals that matter:

- **A builder who prototypes to think.** You would rather show a working thing in two days than write a spec for two weeks, and you learn by building.
- **Architecture-first instincts.** You separate content from workflow, treat knowledge as a single governed source of truth, and design so one thing feeds many channels.

- **Reads search logs as a content backlog.** You treat the questions that came back empty as the next units to write, before you tune ranking.
- **Real experience in regulated, high-volume operations.** You respect legal verbatim and compliance, and you know where a human must stay in the loop.
- **Reasons from the frontline pain.** You start from the member and the advocate, not from the technology.
- **Connects the piece to the whole.** You know which AI use cases wait on this platform and sequence your roadmap around them.

If that describes how you already work, this brief should feel natural. Show us your reasoning and where your past work shaped it.

Show your thinking (weighted as heavily as the prototype)

We are hiring for judgment as much as output. Alongside what you build, make your reasoning visible. The follow-up call will spend most of its time here.

- **The decisions behind the work.** What you considered, what you chose, what you chose not to build, and why.
- **Your assumptions.** What you assumed about the content, the authors, the consumers, the people who search, and the constraints.
- **How your experience shaped this.** Point to one or two things from your past work that influenced your approach. The strongest submissions show what a person has learned elsewhere, applied to this problem.
- **What you would do next.** With more time, and what you would do differently knowing what you know now.

We would rather see a rough prototype you can defend than a polished one you cannot.

How we will evaluate

Dimension	What we are checking
Problem & user insight	Real pain, real users, a clear “why now”
AI that does real work	Extraction, classification, dedup, drift, and retrieval over approved units, scoped by permission, with unanswered questions routed back to authors. A search box over raw files does not score here
Search quality	You measure retrieval before you trust it (a golden set, however small); it cites the unit and says so when it cannot answer
Journey mapping	Current → future → how the product gets there

Execution	The prototype shows author → govern → find and consume, end to end
Judgment	Roadmap order and the chosen tradeoff
Responsible AI	Explicit human approval gate for legal / clinical content
Thinking & accumulated judgment	Clear reasoning, considered alternatives, and prior experience that shapes the approach. Weighted as heavily as the prototype

Guidance and rules

You have 2 to 3 days. Use made-up data only: no real member, provider, or claims data.

Getting knowledge wrong is a compliance and safety issue, so keep a human in the loop. The AI proposes extracted knowledge, classifications, and updates; humans review and publish. Anything legal or clinical must pass an explicit human approval gate before it goes live. Author once; the same approved unit serves each channel and each search.

Search here means retrieval over governed knowledge. It runs on the units you extract and approve, it covers only the knowledge bases the asker may read, and it cites the unit it answers from. If no approved unit supports an answer, it says so. Searching the company's drives and mailboxes is a different product and out of scope.

Be ready to present. The follow-up is a working conversation: we want the reasoning behind each choice, the paths you did not take, and how your experience led you here.

Stretch (optional, for exceptional candidates)

Show the same published knowledge grounding two different channels at once (e.g., advocate guidance and a member chat answer), a drift alert that proposes the fix when a source document changes, or a search-analytics view that finds a blind spot (a question people keep asking that no unit answers) and proposes the unit that would close it.

Use Case 4: HEDIS Gap Closure Automation

Business area: Quality / Stars (HEDIS)

Main AI role: Find evidence in member data that a care gap is already closed (basic). Turn gap data into insight about care quality (intermediate). Predict which members will open a gap and recommend the right intervention (advanced).

You can show only the basic feature, or add one or both of the advanced ones.

This is a self-contained brief. It gives you enough to start. We expect you to research HEDIS, Star Ratings, and quality measurement on your own to build a strong use case.

The problem

HEDIS is the set of quality measures health plans are graded on: screenings, chronic-condition control, medication adherence, and more. A “gap” is a member who is due or overdue for one of these measures with no record that it was done. When a gap closes, the member gets care they needed and the plan earns credit toward its quality scores and Star Ratings, which drive revenue and reputation.

The problem has three layers. First, the data is messy. Proof that care already happened sits in claims, labs, pharmacy records, faxes, and free-text notes, so many “open” gaps are already closed and the evidence never made it into structured data. Today, quality team nurses read through those documents by hand to find and confirm the evidence. That work is slow and expensive, and it does not scale. Second, even with clean data, plans struggle to turn a raw gap list into insight: which measures are slipping, and where gaps cluster by provider, region, or population. Third, most outreach is reactive and generic. Plans send broad reminders late in the year instead of predicting who will fall out of compliance and reaching them early with the right intervention.

Members miss care, nurses waste effort on gaps that were already closed, and every reporting deadline turns into a scramble. Scores end up lower than they should be. You decide how far to take it: evidence only, evidence and insight, or all three.

Who it is for (think about all of them)

- **Quality / HEDIS team nurses** who currently review unstructured documents by hand to find and confirm gap-closure evidence.
- **Care managers and outreach staff** who contact members and help them get care.
- **Providers and their staff** who deliver the care and hold much of the proof that it happened.

- **Members** who are due for care, may not know it, and often face real barriers to getting it.
- **Plan leadership** accountable for Star Ratings and the revenue and reputation tied to them.

What to deliver

Build an AI-powered prototype for this use case and turn in five things.

1. **Problem Brief.** Define the problem, who it is for, and why it is worth solving now. Call out the current pain points and the business opportunity.
2. **Service Design.** Show the current end-to-end user journey and the future one. Explain where your solution fits in that journey. Make the current pain clear and the future improvement obvious.
3. **Working Prototype.** Build a working, interactive demo with a tool like v0, Claude Code, Cursor, Replit, Lovable, Codex, or similar. It does not need to be production ready. It should show the core experience and your prototyping skills.
4. **Product Roadmap.** Lay out a phased plan. For each phase, explain the order and the reason behind it. If you build more than the basic feature, show how evidence, insight, and prediction build on each other over time.
5. **Key Tradeoff.** Name one real tradeoff you made and explain why.

What we are looking for

We care about your product thinking.

- **Problem framing.** Do you understand the real pain, the users, and the business case? Can you explain why now?
- **AI that does real work.** Show how the AI is built into the solution to solve the problem. A basic chatbot wrapper is not enough. The AI should do work that changes the outcome.
- **Journey mapping.** Current journey, then the future end state, then how your product gets there.
- **Execution.** A prototype that clearly shows the core experience.
- **Judgment.** The order of your roadmap and the tradeoff you chose.

Guidance and rules

- You have 2 to 3 days.
- Use made-up or anonymized data only. Do not use real member, provider, or claims data.

- Keep a human in the loop where it matters. Be clear about where the AI decides, where it recommends, and where it hands off to a person.
- Be ready to present and talk through your thinking on a follow-up call.

Use Case 05: Medicare Risk Adjustment Automation

Business area: Risk Adjustment / Medicare Advantage

Main AI role: Read unstructured clinical documents to find and confirm supported diagnoses (HCCs) so risk is captured accurately (basic). Predict undocumented conditions and recommend how to capture them (advanced).

You can show only the basic feature, or add the advanced one too.

This is a self-contained brief. It gives you enough to start. We expect you to research Medicare Advantage, HCC coding, RAF scores, and RADV audits on your own to build a strong use case.

The problem

Medicare Advantage plans are paid based on how sick their members are. Each member's conditions are translated into diagnosis codes, the codes roll up into a risk score (RAF), and that score sets how much the plan is paid to care for that member. Chronic conditions have to be documented again every year. If a condition is not coded this year, it does not count, even if the member clearly has it.

The data is messy. Evidence of a member's conditions is scattered across claims, labs, pharmacy records, faxes, and free-text chart notes. Today, medical coders read through those documents by hand to find and confirm which diagnoses are supported. That work is slow and expensive, and it does not scale. Most capture work is also reactive. Teams chase charts near the submission deadline instead of predicting suspected conditions and prompting the right action before the member's next visit.

Errors hurt in both directions. Under-documentation leaves legitimate revenue on the table and understates how sick the population really is. Unsupported or inflated codes create serious compliance and audit risk (RADV), with paybacks and penalties. You decide how far to take it: surfacing evidence, or also predicting undocumented conditions.

Who it is for (think about all of them)

- **Risk adjustment teams** who run chart review, suspecting, and submissions.
- **Medical coders** who currently review unstructured documents by hand to find and validate supported diagnoses.
- **Providers and their staff** who document conditions and hold much of the proof.
- **Compliance and audit teams** accountable for coding accuracy and RADV exposure.

- **Data and analytics teams** who wrangle the source data and report performance to leadership.
- **Members**, because accurate capture of their conditions drives their care management and the resources set aside for their care.
- **Plan leadership** accountable for revenue accuracy and audit risk.

What to deliver

Build an AI-powered prototype for this use case and turn in five things.

1. **Problem Brief.** Define the problem, who it is for, and why it is worth solving now. Call out the current pain points and the business opportunity.
2. **Service Design.** Show the current end-to-end user journey and the future one. Explain where your solution fits in that journey. Make the current pain clear and the future improvement obvious.
3. **Working Prototype.** Build a working, interactive demo with a tool like v0, Claude Code, Cursor, Replit, Lovable, Codex, or similar. It does not need to be production ready. It should show the core experience and your prototyping skills.
4. **Product Roadmap.** Lay out a phased plan. For each phase, explain the order and the reason behind it. If you build the advanced feature, show how evidence capture and prediction build on each other over time.
5. **Key Tradeoff.** Name one real tradeoff you made and explain why.

What we are looking for

We care about your product thinking.

- **Problem framing.** Do you understand the real pain, the users, and the business case? Can you explain why now?
- **AI that does real work.** Show how the AI is built into the solution to solve the problem. A basic chatbot wrapper is not enough. The AI should do work that changes the outcome.
- **Journey mapping.** Current journey, then the future end state, then how your product gets there.
- **Execution.** A prototype that clearly shows the core experience.
- **Judgment.** The order of your roadmap and the tradeoff you chose.

Guidance and rules

- You have 2 to 3 days.
- Use made-up or anonymized data only. Do not use real member, provider, or claims data.

- Keep a human in the loop where it matters. Confirming a diagnosis and submitting a code are regulated, serious steps. An unsupported or inflated code is a compliance and RADV audit risk. Be clear about where the AI decides, where it recommends, and where it hands off to a person.
- Show your evidence. If your AI suggests a diagnosis or flags a suspected condition, show the source it relied on so a coder or provider can verify it.
- Be ready to present and talk through your thinking on a follow-up call.

Use Case 06: Provider Payment Integrity

Business area: Payment Integrity

Main AI role: Read claims and supporting clinical documents to flag improper payments and policy violations.

This is a self-contained brief. It gives you enough to start. We expect you to research payment integrity, fraud, waste and abuse (FWA), claims coding, and pre-pay versus post-pay review on your own to build a strong use case.

The problem

Payment integrity means making sure claims are paid correctly: the right amount, to the right provider, for services that were actually delivered, covered, and coded properly. Health plans process huge volumes of claims, and a meaningful share of them are wrong: duplicate billing, upcoding, unbundling, services the record does not support, coding errors, or outright fraud, waste, and abuse. Every dollar paid in error is lost, and clawing it back after the fact (“pay and chase”) rarely recovers all of it.

Rules catch the obvious errors. The hard cases hide in unstructured data such as medical records, itemized bills, and notes, which someone has to read against policy and coding rules. Today, claims reviewers, clinical auditors, and investigators do that reading by hand to confirm whether a payment is justified. That work is slow and expensive, and it cannot keep up with claim volume. Most claims are never reviewed at all, and errors slip through.

The plan loses large sums to improper payments and wastes effort chasing recoveries. Providers get frustrated by clumsy audits and takebacks, and legitimate claims get delayed by broad manual review.

Who it is for (think about all of them)

- **Payment integrity / SIU teams** who review claims and pursue improper payments.
- **Claims reviewers and clinical / coding auditors** who currently review records against policy and coding rules by hand.
- **Investigators** who work suspected fraud, waste, and abuse.
- **Providers and their staff** on the receiving end of audits, denials, and takebacks.
- **Data and analytics teams** who build detection logic and report recoveries and leakage.
- **Members**, whose premiums go up and whose trust goes down when money is wasted.
- **Plan leadership** accountable for medical cost, recoveries, and regulatory compliance.

What to deliver

Build an AI-powered prototype for this use case and turn in five things.

1. **Problem Brief.** Define the problem, who it is for, and why it is worth solving now. Call out the current pain points and the business opportunity.
2. **Service Design.** Show the current end-to-end user journey and the future one. Explain where your solution fits in that journey. Make the current pain clear and the future improvement obvious.
3. **Working Prototype.** Build a working, interactive demo with a tool like v0, Claude Code, Cursor, Replit, Lovable, Codex, or similar. It does not need to be production ready. It should show the core experience and your prototyping skills.
4. **Product Roadmap.** Lay out a phased plan. For each phase, explain the order and the reason behind it.
5. **Key Tradeoff.** Name one real tradeoff you made and explain why.

What we are looking for

We care about your product thinking.

- **Problem framing.** Do you understand the real pain, the users, and the business case? Can you explain why now?
- **AI that does real work.** Show how the AI is built into the solution to solve the problem. A basic chatbot wrapper is not enough. The AI should do work that changes the outcome.
- **Journey mapping.** Current journey, then the future end state, then how your product gets there.
- **Execution.** A prototype that clearly shows the core experience.
- **Judgment.** The order of your roadmap and the tradeoff you chose.

Guidance and rules

- You have 2 to 3 days.
- Use made-up or anonymized data only. Do not use real member, provider, or claims data.
- Keep a human in the loop where it matters. Denying a payment, clawing one back, or flagging a provider for fraud are serious, regulated actions. A wrong flag harms a legitimate provider and can create compliance and legal risk. Be clear about where the AI decides, where it recommends, and where it hands off to a person.
- Show your evidence. If your AI flags a claim or an improper payment, show the record and the policy it relied on so an auditor can verify it.
- Be ready to present and talk through your thinking on a follow-up call.

Use Case 07: Clinical Signal

Business area: Clinical Analytics (Inpatient Admission)

Main AI role: Detect inpatient admission hotspots, meaning places where admissions are abnormally concentrated, and correlate signals to find the likely root cause (basic).
Recommend the right intervention and the owner who can act on it (advanced).

You can show only the basic feature, or add the advanced one too.

This is a self-contained brief. It gives you enough to start. We expect you to research inpatient admission drivers on your own to build a strong use case.

The problem

Inpatient (hospital) admissions are the largest and fastest-growing driver of a health plan's medical spend. But a rising admission count tells leaders almost nothing about what to do. The increase could sit in one facility, one member cohort, a bad respiratory season, or an avoidable pattern of care. The same total can come from very different causes, and each one needs a different fix and a different owner.

Today, teams pull reports and spend weeks slicing the data by hand to find where admissions are spiking, guessing at why, and debating who should fix it. By the time they have an answer, the trend has moved and the window to act has closed. Three questions need fast answers: where are admissions abnormally concentrated, what is actually driving them, and who can do something about it. Detecting the hotspot, finding the cause, and making a credible recommendation are each hard on their own.

Avoidable admissions and preventable readmissions cost the plan money. Analysts lose time to manual investigation. Interventions get aimed by intuition rather than evidence, and the same hotspots come back because no one closes the loop from signal to action. You decide how far to take it: detecting and explaining hotspots, or also recommending action.

Who it is for (think about all of them)

- **Medical cost / trend and affordability teams** who monitor inpatient spend and have to explain why it is moving.
- **Clinical analytics and data teams** who currently slice reports by hand to find hotspots and explain them.
- **Care management and clinical operations leaders** who design and run the interventions.

- **Network, contracting, and facility partners**, where cost concentrates and where some of the fixes have to happen.
- **Finance and actuarial leaders** accountable for medical cost trend.
- **Members** affected by avoidable admissions, readmissions, and long stays.
- **Leadership** accountable for affordability and quality of care.

What to deliver

Build an AI-powered prototype for this use case and turn in five things.

1. **Problem Brief.** Define the problem, who it is for, and why it is worth solving now. Call out the current pain points and the business opportunity.
2. **Service Design.** Show the current end-to-end user journey and the future one. Explain where your solution fits in that journey. Make the current pain clear and the future improvement obvious.
3. **Working Prototype.** Build a working, interactive demo with a tool like v0, Claude Code, Cursor, Replit, Lovable, Codex, or similar. It does not need to be production ready. It should show the core experience and your prototyping skills.
4. **Product Roadmap.** Lay out a phased plan. For each phase, explain the order and the reason behind it. If you build the advanced feature, show how detection and root cause feed the recommendation over time.
5. **Key Tradeoff.** Name one real tradeoff you made and explain why.

What we are looking for

We care about your product thinking.

- **Problem framing.** Do you understand the real pain, the users, and the business case? Can you explain why now?
- **AI that does real work.** Show how the AI is built into the solution to solve the problem. A basic chatbot wrapper is not enough. The AI should do work that changes the outcome.
- **Journey mapping.** Current journey, then the future end state, then how your product gets there.
- **Execution.** A prototype that clearly shows the core experience.
- **Judgment.** The order of your roadmap and the tradeoff you chose.

Guidance and rules

- You have 2 to 3 days.
- Use made-up or anonymized data only. Do not use real member, provider, or claims data.

- Keep a human in the loop where it matters. Naming a root cause and choosing an intervention affect clinical care, provider relationships, and budgets. A wrong root cause sends real effort in the wrong direction. Be clear about where the AI decides, where it recommends, and where it hands off to a person.
- Show your evidence. If your AI flags a hotspot, names a cause, or recommends an action, show the signals behind it so a person can check it and trust it.
- Be ready to present and talk through your thinking on a follow-up call.